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How Varying Levels of Knowledge and Motivation Affect Search and Confidence during Consideration and Choice

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Abstract

Using a pull-down menu search technique, we study how ability (knowledge level) and motivation (accountability) interact with decision phase (consideration vs. choice) to affect consumers' search costs (amount of search), benefit (confidence) and efficiency (benefit for a given amount of search). We find that consumers adapt their search strategies across phases and that these adaptations occur in different ways and at different rates for consumers with different levels of knowledge and accountability.

Keywords: knowledge level, accountability, consideration, choice, efficiency

While many of the decisions that we make as consumers are so routine or repetitive that they require little thought or effort, occasionally we face an especially important decision that is complex in terms of the amount of information that can potentially be brought to bear. These "complex" decisions typically include a number of options differing from each other on a number of dimensions. The outcome of such a decision will depend on how much knowledge we bring to the decision, how much we search for relevant information, how we allocate that search effort, and how we evaluate the fruits of our labor.

Research in this area typically considers the costs and benefits of different decision strategies at different phases in the narrowing of options down to a final choice (Hauser and Wernerfelt, 1990; Johnson and Payne, 1984; Roberts and Lattin, 1991). In experimental work, costs are usually assessed by measuring time spent searching and/or number of pieces of information acquired, and benefits are often computed by measuring consumer's attitudes such as confidence. Payne et al. have studied adaptive decision making observing that consumers' tendency to use a repertoire of decision strategies under varying choice conditions is "... response of a limited-capacity information processor to the demands of complex decision tasks" (1993, p. 2). In "deciding how to decide" the decision maker will trade off effort and accuracy, seeking a decision of acceptable quality for an acceptable

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level of effort. The same individual will apply different strategies within the same problem, as that problem evolves. The purpose of the present research is to understand human decision behavior in such an environment by systematically varying task characteristics and individual characteristics and tracking the decision as it evolves. The present study adds to previous research by addressing specific questions concerning the circumstances under which consumers process information with more and less cost (amount of search), with more and less benefit (confidence in quality of selection) and with more and less efficiency (amount of benefit for a given level of search effort).

We examine how ability and motivation impact the process of narrowing choice options to arrive at a final choice. In particular, we manipulate *knowledge level* and consider it to represent the *ability* to process information effectively. We manipulate *accountability* and consider it to represent *motivation* to process information effectively. To observe how decision making changes as the task evolves, we use a two-phase task where consumers are first asked to narrow from a large set of options (in our case, laptop computers) to a smaller set of those they would seriously consider and then to make a final choice from this consideration set. We then focus on the combined effects of knowledge level, accountability and phase, using newly developed measures of search effort and allocation of search effort or strategy use.

Previous research suggests that consumers proceed through different decision phases in different ways (Levin et al., 2000; Levin and Jasper, 1995; Ordonez et al., 1999; Payne et al., 1993). For example, decision makers going through successive phases expend more effort in early phases when forming consideration sets than in later phases when making choices from these sets (Levin et al., 2000, 1998). In terms of strategy, Payne et al. (1993) report that participants when screening options through an alternative-by-attribute matrix begin by using relatively easy to execute non-compensatory attribute-based strategies such as Tversky's (1972) EBA (elimination-by-aspects), but then apply the more cognitively taxing brand oriented WADD (weighted additive alternative-based) strategy to the fewer alternatives that remain. As consumers move through these decision phases, they may become more confident in the quality of their selections because they have inspected more information. This would also mean that they become more efficient across phases because although search decreases, confidence in the quality of the outcome increases.

1. Hypotheses

In this section, we develop specific hypotheses about how knowledge and accountability affect search, search orientation, search efficiency and confidence in the outcome across the two phases of the decision task. We employ a software program called ComputerShop© that provides a rich array of process tracing data for our phased narrowing task. Based on both theoretical considerations and factor analysis (to be described in a later section), we focus on the amount of search, search orientation (alternative-based vs. attribute-based), efficiency and confidence in the outcome as our primary dependent measures.

1.1. Search

In developing our hypotheses about search, we begin by recognizing that across phases consumers generally reduce their effort by reducing the amount of search by dropping out options from further consideration. Regarding ability, initially high knowledge consumers will be especially able to focus and reduce excessive search by using their richly developed knowledge structures with both specific (attribute-level) and abstract information (Alba and Hutchinson, 1987; Spence and Brucks, 1997; Swait and Adamowicz, 2001). Regarding motivation, initially consumers who expect to justify their actions to others will search more while forming consideration sets and making choices than consumers who do not expect to justify their actions (McAllister et al., 1998). However, we expect that these ability and motivation differences will diminish across phases because the consumers who initially expend the most (least) effort will experience the biggest (smallest) drop in effort across phases as they balance total effort against accuracy.

H1: Low knowledge consumers will initially search more than high knowledge consumers, but later this difference will diminish because low knowledge consumers will experience a greater decrease in amount of search.

H2: High accountability consumers will initially search more than low accountability consumers, but later this difference will diminish because high accountability consumers will experience a greater decrease in amount of search.

Additionally, we expect that the amount of search will be influenced by the combination of accountability, knowledge and phase. Based on the assumption that knowledge level taps ability to process information and accountability taps motivation and that these variables combine multiplicatively, the effects of knowledge level should be especially prevalent under conditions of high accountability. Again, however, because of the natural tradeoff of the amount of search across phases, this pattern should be most evident in early phases. Therefore:

H3: The difference in amount of search between high and low knowledge consumers should be greater under high accountability than under low accountability and this pattern should be more evident in the initial phase.

1.2. Search Orientation

Payne et al.'s (1993) concept of adaptive decision making would suggest that consumers in our task will shift search strategies as they move from Phase 1 with many options to inspect to Phase 2 with relatively few options to inspect (see also Libby and Luft, 1993). Based on their analysis, Payne et al. would predict a shift from initial attribute-based search to more alternative based search. Lower knowledge consumers may be slower to shift strategies than higher knowledge consumers because they are not as adept at responding to the changing choice environment. Thus we predict:

H4: Although all consumers may shift their search orientation across phases, high knowledge consumers will shift more quickly than low knowledge consumers.

Regarding motivation, the effect of accountability on search orientation is less clear. On the one hand, literature suggests that consumers who are accountable rely on analytic alternative-based decision making (McAllister et al., 1998; Simonson and Nye, 1992; Tetlock, 1983). On the other hand, strategies that are easier to execute such as elimination-by-aspects should also be easier to explain or justify to others. This would lead to more attribute-based search for high accountability consumers. Because of the conflicting information, we leave as an open-ended research question the effect of accountability on search orientation.

1.3. Satisfaction and Efficiency

We also expect that ability, motivation and phase will affect confidence in selections, and search efficiency, which is operationally defined as confidence level (output) for a given level of search effort (input). All consumers will gain confidence and efficiency as they proceed through the task because consumers can make a more informed decision after they have accessed more of the available information. Regarding ability, high knowledge consumers may be more confident in their selections and be more efficient than low knowledge consumers because high knowledge consumers are more likely to understand what information is important.

H5: Compared to low knowledge consumers, high knowledge consumers will report more confidence and be more efficient.

Regarding motivation, high accountability consumers may be less confident in their selections than low accountability consumers because they will be comparing their decision-making against the people to whom they are accountable rather than against their own well-known internal standards (Tetlock, 1983). This reasoning leads to the following hypothesis:

H6: Compared to low accountability consumers, high accountability consumers will report less confidence in their selections.

1.4. Consideration Sets

High knowledge consumers may form smaller consideration sets than low knowledge consumers because they are better able to discriminate between good and bad alternatives.

Likewise, high accountability consumers may form smaller consideration sets than low accountability consumers because they are more motivated to discriminate between good and bad alternatives.

H7: A) High knowledge consumers will form smaller consideration sets than less knowledgeable consumers and B) high accountability consumers will form smaller consideration sets than less accountable consumers.

2. Method

2.1. Overview

In this 2 (knowledge levels: high vs. low) $\times$ 2 (levels of accountability: high vs. low) $\times$ 2 (phase: consideration vs. choice) experiment, participants select a laptop computer as a gift for a friend. They use a pull-down menu procedure to access as little or as much information as they choose about different options. First, consumers narrow their options to a set that they would like to consider further. Then they make a choice from that set. We manipulate both knowledge level and accountability between subjects by presenting written information prior to the start of the search task. Phase is a within-subject factor.

We used specially developed search monitoring software, ComputerShop©, to present information to the decision-maker with a Windows™-type environment. ComputerShop© presents information in the following way: The computer screen is initially blank, except for menu labels in the upper left corner and three buttons in the upper right. These menus, from which participants access information, are labeled, "Brands," and "Features." Under the Brands main menu, the information is organized by brand and model names in submenus arranged hierarchically. When the participant clicks on the Brands main menu, a menu pulls down that lists the brand names of the computers. When any of these brand names is clicked, a list of the models of that brand name appears. Clicking any of these model names produces a list of the features of the computers. Clicking any of these features reveals the specific value of that attribute for that specific model of notebook computer. The successive menus appear from left to right as the participant progresses through the search, with the exception of the final specific attribute value which appears in the middle of the screen.

Under the Features main menu, the information is organized by attributes. From this menu, participants can compare brands on any of the following attributes: weight, price, CPU type, CPU speed, RAM, screen size, ports and bays, and hard drive capacity. This menu and its submenus are also arranged hierarchically. When the participant clicks on the Features main menu, the list of attributes appears. Clicking on any of these attributes produces a list of the ranges of that attribute. Clicking on any of those ranges reveals a list of model names that fall into that range. When the participant clicks on a model name, a list of attributes appears that allows the participant to determine the exact value of that attribute for that specific model of computer. These menus also appear from left to right across the screen, with the final attribute value appearing in the middle of the screen.

Participants can move between the main menus and among the submenus in virtually any order they choose. ComputerShop records each piece of information that is accessed by each participant; how long the information is viewed, and where it is located in the menu hierarchy. This log file is never seen by the participant.

2.2. *Procedure, Participants and Manipulations*

We randomly assigned groups of consumers to the accountability conditions and within each group, we randomly assigned individual participants to the knowledge conditions. Participants arrived in the computer laboratory in groups of 20 to 25. Once all participants completed informed consent documents, we demonstrated the software and they practiced the experimental task by examining information about soft-sided luggage. Then they read a handout, which presented instructions to manipulate knowledge and accountability, a multiple choice knowledge test and a description of a graduating friend who had taken a new job as an administrative assistant in a commodities brokerage firm. Eight randomly selected participants completed concurrent verbal protocols.

2.2.1. Phase Manipulation In Phase 1 of the experimental task, consumers learned that they should examine the product information and identify laptop notebook computers about which they would seek out more information prior to making a final choice. They learned that in this first phase they could select as many models as they wished, but they were not explicitly informed that they would be making a single choice in a later phase. When they had completed Phase 1, they clicked a button indicating they were ready to move to the next phase. The next screen asked consumers to use a 1 to 7 scale to rate "How confident are you that you have made the best selections?"

The next set of instructions introduced Phase 2 by asking consumers to make a final choice from among the alternatives selected in Phase 1. For this phase, ComputerShop© recompiled the database containing product information. The user interface was exactly as it was in Phase 1 except each participant had available to him or her *only* the product information pertaining to the alternatives selected in Phase 1. After choosing one alternative, respondents indicated a response to the question: "How confident are you that you have made the best selection?"

2.2.2. Knowledge Manipulation In the high knowledge condition, participants received a handout that described in general terms the factors that should be considered when choosing a notebook computer. It also provided a brief description of each of the attributes and described the key differences among three price levels of notebook computers.

2.2.3. Accountability Manipulation In the high accountability condition, participants learned that roughly half of the group would be randomly selected to defend their search strategies to the group. To avoid potential problems of outcome accountability that have been shown in the literature, the handout emphasized that we wanted participants to discuss their overall strategy and not the individual computers that they rejected or selected

(Ordonez et al., 1999). We randomly selected participants in each session to orally defend their strategies.

3. Results

3.1. Manipulation Checks

Scores on the multiple-choice test show that the subjects who were educated prior to performing the experimental task were more knowledgeable about the product class. (Educated: 86% correct, Not Educated: 64% correct; $t_{88} = 4.42$, $p < 0.01$.) Following the experimental task, participants were also asked on a paper-and-pencil survey how likely they thought it was that they would be asked to justify their selection procedures. Those in the accountability condition indicated a significantly higher belief that they would be asked to account for their decision process (High Accountability: 5.38, Low Accountability = 6.79, where 1 = "very likely" and 10 "not likely"; $t_{88} = -3.34$, $p < 0.01$). Both manipulations, therefore, were successful.

3.2. Measures of the Search Process and Factor Analysis

We had nine measures of the search process. To assess the benefits of search, we measure the consumer's confidence in the alternative or alternatives selected at each phase. To measure amount of search, we observe both the time consumers spend arriving at the set (choice) and the number of pieces of information acquired to carry out a decision strategy. To assess how consumers allocate their search effort, we develop measures to assess attention to alternatives and attributes, including time spent on each alternative that they actually inspect, proportion of available alternatives that are examined but not selected (the inept set), number of information acquisitions about each attribute examined (probe) and the average number of attributes inspected per brand examined (breadth). We also assess the extent to which search is alternative-oriented or attribute-oriented by examining each participant's "path" through ComputerShop©'s hierarchical menus. Alternative-oriented search occurs when consumers access information by using the menu labeled "Brands" and its submenus. Attribute-oriented search occurs when consumers access information organized by attributes, using the menu labeled "Features" and its submenus. We measure the proportion of information acquired using each of these main menus and their respective sub-menus.

To reduce these observed variables into a smaller number of created variables, we performed a Factor Analysis. These nine measures of search were subjected to a principal component analysis using ones as prior communality estimates. The principal axis method was used to extract the components and this was followed by a direct oblimin (non-orthogonal) rotation. The non-orthogonal rotation was considered more theoretically sound given the inter-relatedness of the measures. Only the first three components displayed eigenvalues greater than 1 and the results of a scree test also suggested that only the first three components were meaningful. Combined, these three factors accounted for a total of 75.07% of the variation among the nine measures of search. The following dependent measures loaded on what we called the Search factor at a value of at least 0.40 (absolute value): total acquisitions (0.89), total time (0.74), probe (0.86), inept set (0.72) and time per alternative (−0.81). Attribute-oriented acquisitions (0.94), breadth (attributes examined per alternative) (0.65) and brand-oriented acquisitions (−0.98) loaded most strongly on the second factor, Attribute Orientation. Because confidence was the only dependent measure that loaded on the third factor at a level greater than 0.40, we labeled this factor Confidence. We then performed three univariate analysis of variance procedures using Effort, Attribute Orientation and Confidence as the dependent variables and all of the independent variables in mixed models.

3.3. Search

We present Analysis of Variance results in Table 1 and the means for the Search factor are displayed in Figure 1. As predicted, consumer search decreased from Phase 1 to Phase 2. Additionally, ability and motivation operate as predicted in H1 and H2. High knowledge consumers exhibited a lower level of search than low knowledge consumers and low accountability consumers exhibited a lower level of search than high accountability consumers. Follow-up t-tests revealed that consistent with H1, the difference in search between high and low knowledge consumers was significant in Phase 1 ($t_{131} = -2.5$, $p = 0.02$), but not in Phase 2 ($t_{131} = -0.68$, $p = 0.50$). Consistent with H2, the difference between high and low levels of accountability was significant in Phase 1 ($t_{131} = 3.11$, $p = 0.00$), but not in Phase 2 ($t_{131} = 0.51$, $p = 0.61$). Consistent with H3, there is a significant three-way interaction between phase, accountability and knowledge on search ($F_{1,252} = 3.87$, $p < 0.05$) (see Figure 1). As predicted, the difference in the amount of search in Phase 1 between high and low knowledge consumers was significant under high accountability ($t_{131} = -2.89$, $p = 0.00$), but not under low accountability ($t_{131} = -0.68$, $p = 0.49$). In Phase 2 neither difference was significant. In sum, the data on search were consistent with all three hypotheses, H1, H2 and H3.

Table 1. ANOVA Results

	Dependent variables							
	Effort		Attribute orientation		Satisfaction		Consideration set size	
	F	$p<$	F	$p<$	F	$p<$	F	$p<$
Between subjects effects								
Accountability (A)	6.71	0.01	13.7	0.00	0.01	0.91	3.13	0.08
Knowledge (K)	4.89	0.03	0.01	0.93	9.02	0.01	0.61	0.44
A × K	0.02	0.89	1.18	0.28	0.171	0.68	0.09	0.76
Within subjects effects								
Phase (P)	194.6	0.00	23.4	0.01	15.8	0.00	–	–
P × A	4.5	0.04	0.10	0.75	0.04	0.85	–	–
P × K	2.26	0.13	1.49	0.31	0.86	0.35	–	–
P × A × K	3.87	0.05	0.01	0.92	0.02	0.90	–	–

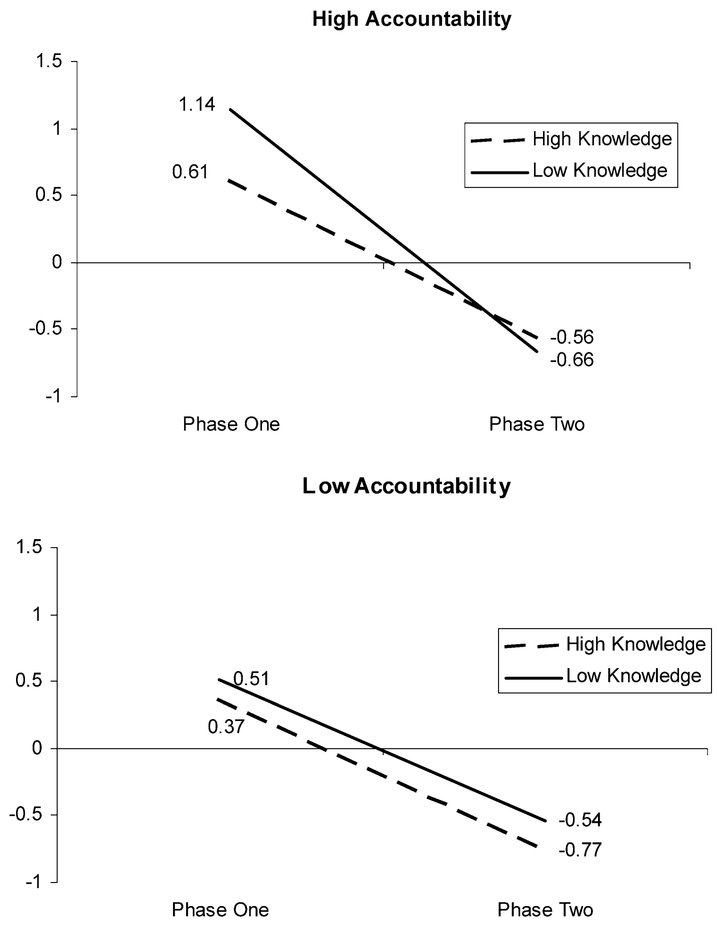

Figure 1. Changes in Amount of Search across Phases for each Combination of Knowledge and Accountability.

3.4. Search Orientation

Perhaps the most unexpected and therefore the most interesting result involves search orientation. Table 1 shows that consistent with H4, consumers adapted their search orientation when moving across phases. However, unexpectedly, consumers significantly increased rather than decreased attribute oriented search (Phase 1: -0.26 vs. Phase 2: 0.22). This increase in attribute search may occur because at the choice phase, consumers could screen the subset of acceptable alternatives selected in Phase 1, by using the pull-down attribute menus. For example, one consumer's verbal protocol stated that at the choice phase he used price to choose between a Dell and Hewlett–Packard: "By just looking at the prices, I would probably choose the Dell because it has a three-year warranty and the price is a lot better than the Hewlett Packard."

A number of factors have been shown to influence decision makers to use relatively more attribute-oriented or alternative-oriented search. These include the number of alternatives, the number of attributes about which each alternative is described, the degree of similarity among the choice options, and time pressure (Payne et al., 1993). Shifts in strategy from more attribute-based strategies early in the decision to more alternative-based strategies later have been observed in verbal protocols (Bettman and Park, 1980; Payne, 1976). However, Bettman and Kakkar (1977) found that the structure of the information presented has a significant impact on a consumer's choice of processing strategy. Consumers appear to process information using the strategy that is easiest to execute given the way the information is displayed. It is likely that participants gaining experience with the pull-down menu procedure learned the ease of searching by attributes and increased their use of this strategy over time. This result is consistent with earlier findings in which participants performing a series of three search tasks and using the current pull-down menu procedure to examine different sets of alternatives increased their use of attribute-oriented search across the three tasks (Huneke, 1996). Thus, in keeping with Bettman et al., our participants were adaptive and, as will be shown shortly, were able to increase search efficiency across phases.

Regarding ability, H4 also predicted that high knowledge consumers would shift their strategy more quickly than low knowledge consumers. However, knowledge did not affect search orientation and the interaction between knowledge and phase was not significant. In response to our research question about the effects of motivation (accountability) on search orientation we find that accountable participants were less apt than those not accountable to use attribute based search (HA -0.24 vs. LA 0.20), supporting the view that consumers who are accountable are more apt to rely on analytic alternative based decision making (Tetlock, 1983). Thus we find evidence that all consumers adapted their search orientation to the different phases, but that neither ability nor motivation affected how quickly consumers adapted.

3.5. Confidence and Efficiency

Figure 2 and Table 1 show that, consistent with H5, confidence increases across phases. While it should also be noted that this finding may not generalize to choice situations where unfavorable values on specific attributes present the decision maker with an emotion-laden choice (see, for example Luce et al., 2000, for discussion of emotion in consumer decision making), high knowledge consumers, with a better understanding of what constitutes an optimal choice, experience greater confidence than low knowledge consumers, and this difference persists across phases. Regarding motivation and confidence (H6), we found that accountability did not impact consumer confidence in either phase. Perhaps confidence is based on attaining internal standards, independent of whether one has to account to others.

In addition, the previously reported findings of reduced search and increased confidence over phases indicate that consumer efficiency increases across the task. Because confidence was self-rated, our measure focuses on perceived efficiency, which may be a surrogate for

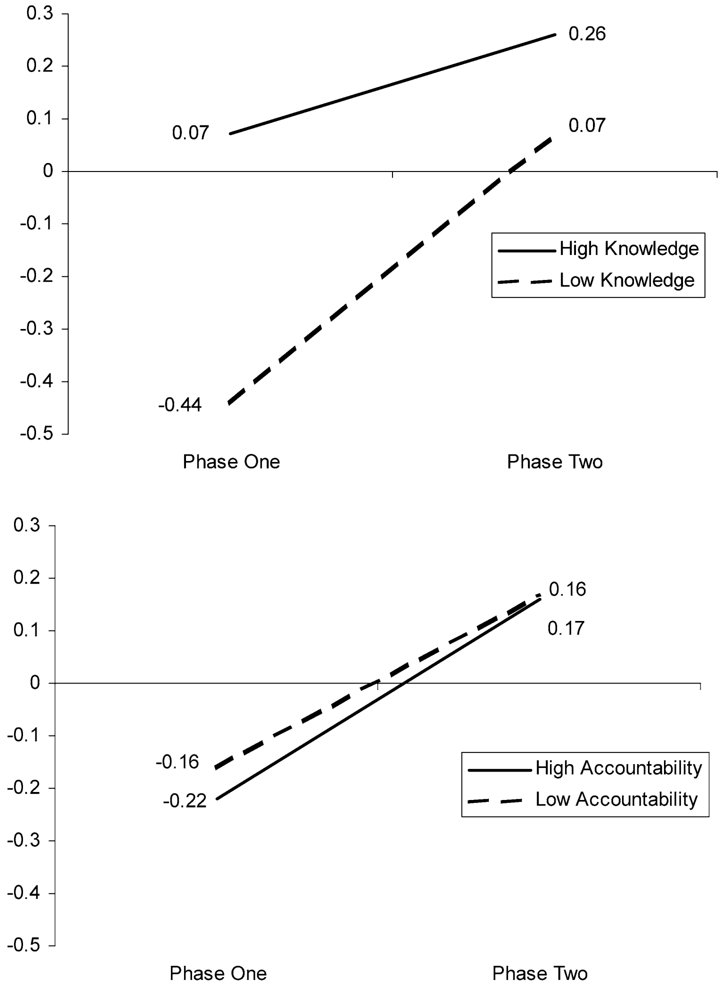

Figure 2. Changes in Confidence across Phases for each Combination of Knowledge and Accountability.

satisfaction. We analyzed the results for efficiency separately in each phase. Using ANCOVA, we conducted an analysis using the Confidence factor as the dependent variable, accountability and knowledge and their interactions as the independent variables and the Search factor as a covariate with data from the first phase. Consistent with H5, we find that knowledge has a significant effect on perceived efficiency ($F_{1,126} = 7.5$, $p < 0.01$), such that high knowledge consumers are more efficient than low knowledge consumers. In the ANCOVA analysis of Phase 2, the Knowledge variable does not achieve significance ($F_{1,126} = 2.05$, $p = 0.15$), indicating a reduced effect of knowledge on perceived efficiency in Phase 2, where all participants are working with a reduced set of alternatives.

3.6. Consideration Sets

In partial support of H7, there was a marginally significant effect of accountability on consideration set size, such that those who are accountable form smaller consideration sets than those who are not accountable (HA 3.37 vs. LA 3.9); no other effects are significant.

4. Conclusions

The results from this study both support earlier research and theory and provide new insights about the behavioral decision processes related to the narrowing of options and about the factors that impact these processes. In support of Payne et al.'s (1993) cost-benefit analysis of "adaptive" decision making, we show that consumers adapt their decision making when first forming a consideration set and then making a choice. They reduce amount of search, change their search orientation, become more confident in their selections and increase efficiency. We also show that groups with different levels of ability adapt their decision making strategies in different ways to changing task demands. For example, high knowledge consumers when initially faced with a large number of alternatives searched less and searched more efficiently than those with lower prior knowledge. All consumers were able to increase efficiency across phases as a result of narrowing their options. Interestingly the observed shift in search orientation across stages applied to both knowledge groups. Strategy choice seems to be more a matter of personal style than level of knowledge (see Levin et al., 2000 for the role of need for cognition).

Consumers who were made to feel accountable to others for the manner in which they arrived at their choices searched more in the process of narrowing choice options (Phase 1), consistent with the assumption that accountability has a motivating influence on consumer behavior. One way to conceptualize this is that accountable consumers are more involved. Further, accountability prompted a more complex (alternative-oriented) search process and, despite their accountability to others, they were no less confident in the outcome of their search.

Our research clearly shows that consumers faced with the complex task of considering and then choosing between a set of multi-attribute alternatives are influenced by how much knowledge they bring to the task and their motivation or involvement in the task. Future research may want to explore how other consumer characteristics influence the costs consumers are willing to incur in the process of narrowing choice options and the benefits that result.

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